MZUZU UNIVERSITY

Department of Mathematics

EMT 1201 - Calculus I

End of Semester Supplementary Examination

Date **Time: (3 hours)**

1. Check that the paper has **12 questions**.
2. Attempt all questions
3. Write answers in the space provided
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

Name _____

Reg Number , Programme: _____

Professor's / Lecturer's name: _____

1. Evaluate each of the following (giving reasons)

   (a) $\cos 5 + \cos(5 - 3\pi)$ (3)

   Solution: $\cos 5 + \cos(5 - 3\pi)$
   $= \cos 5 + \cos(5 - \pi)$ (since cos has period 2π)
   $= \cos 5 - \cos 5$ (since $\cos(\theta - \pi) = -\cos\theta$)
   $= 0$

   (b) $\tan 1 + \tan(-1 + 19\pi)$ (3)

   Solution: $\tan 1 + \tan(-1 + 19\pi)$
   $= \tan 1 + \tan(-1)$ (since tan has period π)
   $= \tan 1 - \tan 1$ (since $\tan(-\theta) = -\tan\theta$)
   $= 0$

2. Show that the following are identities

   (a) $\tan\theta(1 - \cot^2\theta) + \cot\theta(1 - \tan^2\theta) = 0$ (3)

   Solution: $\tan\theta(1 - \cot^2\theta) + \cot\theta(1 - \tan^2\theta)$
   $= \frac{\sin\theta}{\cos\theta}\theta(1 - \frac{\cos^2\theta}{\sin^2\theta}) + \frac{\cos\theta}{\sin\theta}\theta(1 - \frac{\sin^2\theta}{\cos^2\theta})$
   $= \frac{\sin\theta}{\cos\theta} - \frac{\cos\theta}{\sin\theta} + \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta}$
   $= 0$

   Alternatively
   $\tan\theta(1 - \cot^2\theta) + \cot\theta(1 - \tan^2\theta)$
   $= \tan\theta - \tan\theta \times \frac{1}{\tan^2\theta} + \frac{1}{\tan\theta} - \frac{1}{\tan\theta} \times \tan^2\theta$
   $= \tan\theta - \frac{1}{\tan\theta} + \frac{1}{\tan\theta} - \times \tan\theta$
   $= 0$

   (b) Prove that $(\sin\theta - \cos\theta)^2 = 1 - \sin 2\theta$ (2)

   Solution: LHS $= (\sin\theta - \cos\theta)^2 = \sin^2\theta - 2\sin\theta\cos\theta + \cos^2\theta$ Then $\sin^2\theta + \cos^2\theta = 1$ so we have $1 - 2\sin\theta\cos\theta$ which is $1 - \sin 2\theta$

3. (a) Specify the period and amplitude for the function in Figure 1 (2)

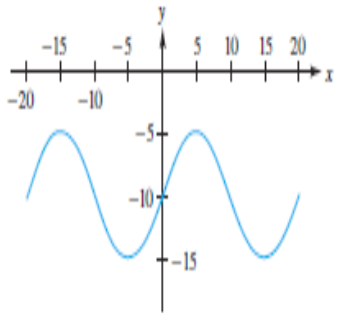


Figure 1:

Solution: The amplitude of the function is 5 and its period is 20.

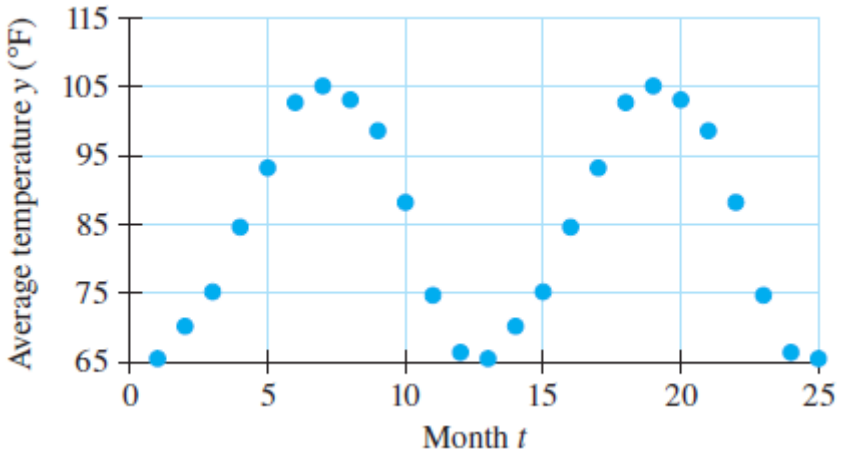
(b) State whether the functions $f(\theta) = \sin\theta$ and $g(x) = \cos\theta$ are increasing or decreasing on the interval $\frac{5\pi}{2} < \theta < \frac{7\pi}{2}$. (2)

Solution: $\frac{5\pi}{2} < \theta < \frac{7\pi}{2}$, $\sin\theta$ is decreasing. But $\cos\theta$ is decreasing between $\frac{5\pi}{2}$ and 3π and is increasing between 3π and $\frac{7\pi}{2}$.

(c) Graph the function $y = 1 - \cos(2x - \frac{\pi}{3})$ for one period. Specify the amplitude, period and the x-intercepts. (5)

Solution: The amplitude is 1, the graph is centred on $y = 1$. Furthermore, its a reflection of the standard $\cos x$ on the x-axis. The period is $\frac{2\pi}{2} = \pi$ and the phase-shift is $\frac{\pi}{3}/2 = \frac{\pi}{6}$.

4. The following scatter plot displays average temperature data for a location north of the Arctic Circle. Use it to find a periodic function of the form $y = A\sin(Bt - C) + D$ whose values approximate the monthly temperatures for the location. (6)



Solution: From the plot, we note that $D = 85$ and amplitude is 20.
Now period is 16 so that $\frac{2\pi}{B} = 16$. Thus $B = \frac{\pi}{8}$.
Phase-shift is approximately 4 so that $\frac{C}{B} = 4$. Therefore, $C = \frac{\pi}{2}$.
So, the function is $y = 20\sin(\frac{\pi t}{8} - \frac{\pi}{2})$.

5. Determine all the solutions in the interval $(0°, 360°)$

   (a) $\sin 2\theta = -2\cos\theta$ (3)

   Solution: $\sin 2\theta = -2\cos\theta$
   $\Rightarrow 2\sin\theta\cos\theta + 2\cos\theta = 0$
   $\Rightarrow (\sin\theta + 1)\cos\theta = 0$
   $\Rightarrow \sin\theta = -1$ or $\cos\theta = 0$
   $\Rightarrow \theta = 270°$ or $\theta = 90°$ or $270°$
   so $\theta = 90°$ or $270°$

   (b) $\sin 3x = 1$ (3)

   Solution: We know that $\sin 90° = 1$
   So $3x = 90$
   General solution given by $3x = 90 + 360k \Rightarrow x = 30 + 120k$
   And so all solutions between 0 and 360 are: $30°, 150°, 270°$

6. Without using a calculator evaluate the following:

   (a) $\sin^{-1}\frac{\sqrt{3}}{2}$ (2)

   Solution: Looking at the 30-60-90 traingle it is immediate that $\sin^{-1}\frac{\sqrt{3}}{2} = 60°$

   (b) $\tan\sin^{-1}\frac{20}{29}$ (3)

   Solution: Using pythagoras, there is a 20-21-29 right-angled triangle. Using this triangle it is easy to conclude that $\tan\sin^{-1}\frac{20}{29} = \frac{20}{21}$

   (c) $\sec(\cos^{-1}\frac{\sqrt{2}}{2} + \sin^{-1}(-1))$ (3)

   Solution: $\sec(45 + 270) = \sec(315) = \sec(45) = \frac{1}{\cos(45)}\sqrt{2}$

7. (a) Let (3)

$$f(x) = \begin{cases} 5 - x & \text{if } x < 4 \\ \frac{1}{5-x} & \text{if } x \geq 4 \end{cases}.$$

   Determine if $f(x)$ is continuous at $x = 4$

Solution: $\lim_{x\to 4^-} f(x) = \lim_{x\to 4^-} 5 - x = 5 - 4 = 1$ and $\lim_{x\to 4^+} f(x) = \lim_{x\to 4^+} \frac{1}{5-x} = \frac{1}{5-4} = 1$
Hence $\lim_{x\to 4} f(x) = 1$. Now $f(4) = \frac{1}{5-4} = 1$. Since the limits exists and it is equal to the value of f, then f is continuous at 4.

(b) The cost of food production is thousands Kwacha at a restaurant after t months is given by $C = f(t) = 10t^3 - 10t$.

i. Use the formula $f'(t) = \lim_{h\to 0} \dfrac{f(t+h) - f(t)}{h}$ to find an expression for the rate of change in the food production cost with respect to the number of months elapsed. (3)

Solution: The expression is $f'(t) = \lim_{h\to 0} \dfrac{f(t+h) - f(t)}{h}$
$= \lim_{h\to 0} \dfrac{10(t+h)^3 - 10(t+h) - 10t^3 + 10t}{h}$
$= \lim_{h\to 0} \dfrac{10t^3 + 30t^2h + 30th^2 + 10h^3 - 10t - 10h - 10t^3 + 10t}{h}$
$= \lim_{h\to 0} \dfrac{30t^2h + 30th^2 + 10h^3 - 10h}{h}$
$= \lim_{h\to 0} 30t^2 + 30th + 10h^2 - 10 = 30t^2 - 10$

ii. Use the expression in part (a) to find the rate of change in the production costs at 3 months (1)

Solution: $f'(3) = 30 \times 3^2 - 10 = 260$ thousands kwacha

8. Differentiate (5)

(a) $g(u) = \sqrt{2}u + \sqrt{3u}$ (2)

Solution: $g'(u) = \sqrt{2} + \frac{3}{2}(3u)^{-\frac{1}{2}}$

(b) $f(x) = xe^{x^2} \csc x$. (5)

Solution: We apply the Product and Chain Rules to get the derivative. Thus

$$\begin{aligned} f'(x) &= e^{x^2} \csc x + x \times 2xe^{x^2} \csc x - xe^{x^2} \times \csc x \cot x \\ &= e^{x^2} \csc x(1 - x \cot x + 2x^2) \end{aligned}$$

(1)

9. Find the equation of the tangent line to the curve $f(x) = 1 + e^{-x}$ at the point $P(0, 1)$. (4)

Solution: Using the Chain Rule we get

$$\begin{aligned} f'(x) &= e^{-x} \times -1) \\ &= -e^{-x} \end{aligned}$$

(3)

so that $m = f'(0) = -1$. Equation of tangent line to $f(x)$ is $y - 1 = -1(x - 0)$ or $y = 1 - x$

10. Find the first and second derivatives of $g(t) = e^{at} \sin bt$. (6)

Solution: First derivative of $g(x)$ is given by

$$\begin{aligned} g'(x) &= ae^{at} \sin bt + e^{at} \times b \cos bt \\ &= ae^{at} \sin bt + be^{at} \cos bt \end{aligned}$$

[2 marks]

and its second derivative, $g''(x)$ is given by

$$\begin{aligned} g''(x) &= a^2 e^{at} \sin bt + abe^{at} \cos bt + abe^{at} \cos bt - b^2 e^{at} \sin bt \\ &= (a^2 - b^2) e^{at} \sin bt + 2abe^{at} \cos bt \\ \textbf{OR} \quad &= e^{at}[(a^2 - b^2) \sin bt + 2ab \cos bt] \end{aligned}$$

[4 marks]

11. Differentiate the functions

(a) $4 \cos x \sin y = 1$ (3)

Solution:

$$\begin{aligned} 4 \cos x \cos y \frac{dy}{dx} + \sin y(-\sin x) &= 0 \\ \cos x \cos y \frac{dy}{dx} &= \sin y \sin x \\ \frac{dy}{dx} &= \frac{\sin y \sin x}{\cos y \cos x} \\ &= \tan x \tan y \end{aligned}$$

(b) $x^2 + xy - y^2 = 4$ (3)

Solution:

$$2x + x\frac{dy}{dx} + y - 2\frac{dy}{dx} = 0$$
$$(x-2)\frac{dy}{dx} = -(2x+y)$$
$$\frac{dy}{dx} = -\frac{2x+y}{x-2}$$

(c) $y = \ln(x + \sqrt{1+x^2})$ (3)

Solution: Let $u = x + \sqrt{1+x^2}$. Then $\frac{du}{dx} = 1 + \frac{x}{\sqrt{1+x^2}}$. We have $y = \ln u$ and $\frac{dy}{du} = \frac{1}{u}$. Hence $\dfrac{dy}{du} = \dfrac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}}$

12. Find the critical numbers of $f(x) = x\sqrt{x - x^2}$. (5)

Solution:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{1}{2}(x-x^2)^{-\frac{1}{2}} \cdot \frac{d}{dx}(x-x^2) + \sqrt{x-x^2} \\ &= \frac{x(1-2x)}{2\sqrt{x-x^2}} + \sqrt{x-x^2} \\ &= \frac{x - 2x^2 + 2x - 2x^2}{2\sqrt{x-x^2}} \\ &= \frac{-4x^2 + 3x}{2\sqrt{x-x^2}}\end{aligned}$$

Critical values occur where the derivative is 0 or where it is undefined. When the derivative is 0, we get $x(-4x+3) = 0$. Hence $x = 0$ and $x = \frac{3}{4}$. The derivative is undefined for $2\sqrt{x-x^2} = 0$ or $x - x^2 < 0$. This gives $x = 0$ and $x < 0$ or $x > 1$. The critical values of the function are hence $x = 0$, $x = \frac{3}{4}$ and $x = 1$.